

Deciding what to be updateful about

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September 15, 2023

Any solution to the foundational problem has to (more explicitly or more implicitly) decide what to be updateful about. I suspect that the idea here presented (being very explicit) is the natural way to do it. Conceptually, it makes sense that what to be updateful about is context-dependent, and since the agent's knowledge of the context is stored in its beliefs, it's to be expected that we need to explicitly check its beliefs to decide.

Imagine that, when deciding what action to take on seeing observation $\mathcal{Q}$, instead of argmaxing a on $\mathbb{E}^{\mathbb{P}}(U|\mathbf{A}(\mathcal{Q}) = a)$ (not updating at all) or on $\mathbb{E}^{\mathbb{P}}(U|\mathbf{A}(\mathcal{Q}) = a \wedge \mathbb{Q}_m = \mathcal{Q})$ (updating completely), we choose some different sentence $\sigma(\mathcal{Q})$, and use $\mathbb{E}^{\mathbb{P}}(U|\mathbf{A}(\mathcal{Q}) = a \wedge \sigma(\Sigma, \mathcal{Q}))$. This $\sigma(\Sigma, \mathcal{Q})$ will be

$$\bigwedge_{\varphi \in \Sigma \cap \text{Support}(\mathcal{Q})} \mathbb{Q}_m(\varphi) = \underline{\mathcal{Q}(\varphi)}$$

for some set Σ of logical sentences. That is, we are updateful exactly about the sentences in Σ (and that, of course, $\mathcal{Q}$ has opinions on).

Our infinite prior $\mathbb{P}$ has some beliefs about the consequences of policies. Some are simple, like

$$\forall \mathcal{Q} (P(\mathcal{Q}) \wedge \mathbf{A}(\mathcal{Q}) = a \rightarrow \uparrow U)$$

But it also has some beliefs about more elaborate policies. Consider the policy we'd obtain if we were updateful exactly about the sentences in Σ , as formalized above. This policy can be described in our logical language, say by a logical formula $\text{Policy}(\Sigma, \mathcal{Q})$ ¹ (for infinite Σ , say they are described by a finite program that decides membership). A $\mathbb{P}$ obtained from an advanced enough $\mathbb{Q}_k$ will have opinions about these policies, for example of the form

$$(\forall \mathcal{Q} (\mathbf{A}(\mathcal{Q}) = \text{Policy}(\Sigma, \mathcal{Q})) \rightarrow \uparrow U)$$

That is, "if I always update on exactly Σ , I'll do well".

There are many equivalent ways to represent these sentences (opinions about Policy_Σ), but we just fix one syntactic shape to check, as always.

So, once we have $\mathbb{P}$ (and before taking any further actions), we find the Σ that argmaxes

$$\mathbb{E}^{\mathbb{P}}(U|\forall \mathcal{Q} (\mathbf{A}(\mathcal{Q}) = \text{Policy}(\Sigma, \mathcal{Q})))$$

¹We could also have Σ not be a variable, and instead have one formula $\text{Policy}_\Sigma()$ for each Σ . But maybe having it be an explicit parameter makes it easier for the Inductor to quickly notice some patterns.

We fix this Σ , and proceed as usual, but adding the corresponding $\sigma(\Sigma, \mathcal{Q})$ to our conditionals.

This is still too coarse, since we just go with the same Σ set for eternity. But we can straightforwardly improve $\text{Policy}()$, and have a more complicated object instead of Σ , that allows for things like "before round n use Σ_1 , and after Σ_2 ". More interestingly, we can accommodate for several meta-levels of deciding which information to update on: "first check whether $P(\mathcal{Q})$, and if that's the case check for $S(\mathcal{Q})$, otherwise don't check anything else". That is, intricately dealing with info-hazards.

Addendums:

Might there be a way to have, or approximate, ω meta-levels of deciding which information to update on? That would resonate with "the agent delegates to itself".

Well, consider the following $\text{Policy}()$. When $\mathbb{P}$ is frozen, it only considers 1 meta-level up ("which things should I update on, and how often"), as above. But at some finite time, it starts considering the 2nd meta-level ("how often, and how, should I be thinking about what to update on, and which updated knowledge should I bring to this reflection"). From there on, this 2nd meta-level is fixed, and the 1st meta-level is updated according to the 2nd meta-level's opinions. For example, maybe the 2nd meta-level is worried that some parts of the tree are very updateful and others very updateless, and so recommends re-considering how much to update (that is, re-writing the 1st level) very often. And so on until the 3rd meta-level kicks in, etc.²

$\mathbb{P}$ could have the opinion that the above $\text{Policy}()$ is good. Of course, the above $\text{Policy}()$ doesn't pertain to any finite meta-level (it is ω -level), but it can be described by a program, or equivalently a formula. Maybe we can make the Σ object so complex, that it can describe Policies as complex as the above. That will look more and more like letting the Σ be arbitrary halting computations.

As I've mentioned in the past, this just seems like a fancier version of Policy Selection. I've felt so many natural pulls in this direction (when trying to approximate "non-catastrophic updateless behavior"), I've felt so many "whispers from the math", that I think we should be okay with that. Nonetheless, I remain excited for any more natural way to implement the ω levels. I do think the implementation will always necessarily be "potentialist" (increasing a finite hierarchy instead of literally having an infinite hierarchy, as in Nisan's thing), since ultimately our prior at finite times only has a finite amount of information, so it wouldn't make sense to ask its opinion on arbitrarily complex sets of sentences. But even a potentialist implementation maybe could stem from a "more insightful core", as opposed to the obvious and explicit implementation of the meta-levels above. Unfortunately, I don't have any ideas for such an implementation yet.

²As written, when we reach the n th meta-level for the first time, the opinions on it are completely non-updated (and the below levels will update according to these opinions). We probably want to change this, by stating they should have updated a bit (even the first time they are considered), so that we get learning as we wanted.